What momentum says about impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect stopping impulse to average force without claiming a complete damage forecast.

SUCCESS CRITERIA - I can...

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Equal-mass carts have velocities **+5** and **-3 m/s** before an ideal **1** D elastic collision. **Give both final velocities**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

Worked reasoning: Two incoming objects have equal momentum **12 kg·m/s** but masses **2** and **4 kg**, so their speeds are **6** and **3 m/s**. Momentum alone does not name every possible consequence of stopping them.

Your turn: Draw equal momentum arrows and unequal velocity arrows.

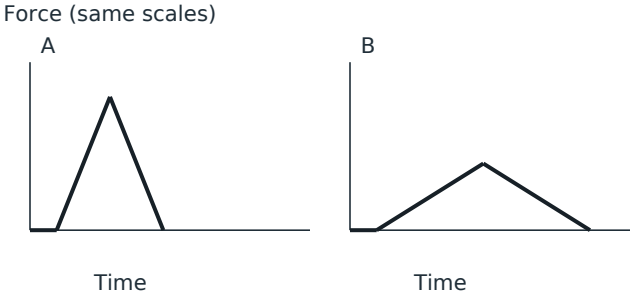


2 kg; 6 m/s 4 kg; 3 m/s

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal-momentum objects stop in **0.1 s** and **1 s**. Which has greater average force magnitude, and by what factor?



Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical **10 kg** target moves at **4 m/s** and stops in **4 s**. **Complete its momentum row**. Then **find the magnitude of the** average stopping force. **State the force direction**.

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

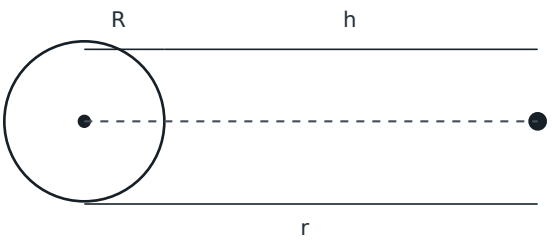
05 Explain your reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A planet has surface field **144 N/kg**. At center distance **6** R, find field and the force on a **8 kg** object.
Label the center distance, radius and field direction.



Marzano check - circle: ① ② ③

07 Exit check

If mass doubles while speed and stopping time stay fixed, how do momentum and average stopping force change?

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. Both objects stop in **0.5 s**.
1. **Compare their average stopping forces**.
2. **Explain why this** does not establish identical peak forces or damage.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③